

DEENA LAD

ML Research Engineer · Spatio-Temporal AI · Climate & Scientific ML

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PROFESSIONAL SUMMARY

M.Tech Gold Medalist (GPA 9.61) and ML Research Engineer with hands-on production experience building end-to-end deep learning pipelines on multi-terabyte scientific datasets. Currently co-authoring a KDD 2026 paper on neural weather forecasting at IIT Gandhinagar's Sustainability Lab. Proven track record deploying spatio-temporal models (U-Net, Transformers, ConvLSTM) that outperform physics-based baselines. Seeking ML/AI Engineering or Applied Research roles where I can turn cutting-edge research into scalable, real-world systems.

TECHNICAL SKILLS

Languages & Data: Python (advanced), SQL, Bash; NumPy, Pandas, Regex, Feature Engineering, Dataset Curation

Deep Learning & ML: PyTorch (primary), TensorFlow, Scikit-learn, PyTorch Lightning; CNNs, LSTM/GRU, ConvLSTM, U-Net, Transformers, YOLO

Generative AI & NLP: Hugging Face Transformers, LangChain, LLM fine-tuning, Prompt Engineering, BERT, Text Classification

Scientific / Geospatial: NetCDF, HDF5, xarray, h5py, GeoPandas, PyProj; WRF & ERA5 reanalysis, INSAT satellite data, Spatio-temporal modeling

MLOps & Deployment: Docker basics, Flask/REST APIs, Git, GPU cluster training (SLURM), Weights & Biases, Hyperparameter Optimization, CI/CD basics

Visualization & Cloud: Matplotlib, Seaborn, Plotly, AWS basics

PROFESSIONAL EXPERIENCE

Junior Research Fellow

Sustainability Lab, IIT Gandhinagar (PI: Prof. Nipun Batra)

Aug 2025 – Present

Gandhinagar, IN

- Architected an end-to-end ML pipeline ingesting 500 GB WRF simulations and 2+ TB atmospheric reanalysis data, including scalable NetCDF loading, distributed preprocessing, and GPU-accelerated training.
- Designed and trained spatio-temporal deep learning models (U-Net + FiLM conditioning, Transformer architectures) for 3–24 hour weather forecasting, achieving $RMSE \approx 2.4^\circ C$ for temperature—surpassing physics-based NWP baselines in inference speed by $10\times$.
- Co-authoring a research paper submitted to KDD 2026 (AI for Science Track); mentored 3 students on ML best practices and geospatial data engineering.
- Built modular, reusable ML components data loaders, training harness, evaluation suite.

Research Intern - Deep Learning for Satellite Imagery

Space Applications Centre, ISRO

Jul 2024 – Apr 2025

Ahmedabad, IN

- Developed CNN and ConvLSTM models for tropical cyclone intensity estimation and next-frame forecasting using 1.5+ TB INSAT-3D HDF5 data (2013–2023, 30-min resolution) over Indian Ocean storms.
- Engineered end-to-end preprocessing and labeling pipelines using IMD track data and Advanced Dvorak Technique (ADT) software, curating 5K annotated images expanded to $\sim 10K$ via class-balanced augmentation.
- Delivered spatio-temporal storm-evolution models, demonstrating transfer of academic DL techniques to operational satellite analysis.

ML/Data Engineering Intern

Jio Platforms Limited, Reliance

May 2024 – Jul 2024

Remote

- Built scraping and ETL pipelines to curate multi-modal sports video datasets from YouTube and sports platforms for athlete personality analysis (Big Five traits).
- Applied prompt engineering via Groq API to generate structured LLM annotations at scale, producing labeled datasets used to benchmark the team's personality-prediction models.
- Demonstrated applied use of generative AI tooling (LLMs, structured output prompting) in a production research workflow.

SELECTED PROJECTS

Neural Weather Forecasting — KDD 2026 Submission | *PyTorch, U-Net, FiLM, xarray, HPC*

- Research-grade pipeline for atmospheric variable prediction over 3–24 hour horizons using ERA5 and WRF data; $RMSE \approx 2.4^\circ C$, with $10\times$ faster inference than WRF physics-based model.

Tropical Cyclone Analysis — INSAT-3D (ISRO) | *PyTorch, ConvLSTM, HDF5, OpenCV*

- Dual-model system: CNN for real-time intensity estimation + ConvLSTM for storm-trajectory next-frame prediction over Indian Ocean cyclone dataset (1.5 TB, 10 years).

Thoracic Disease Detection — Pediatric Chest X-Ray | *Python, YOLOv8/v10, TensorFlow, DICOM*

- Object detection pipeline on VinDr-PCXR (9,125 studies, 36 findings) with DICOM parsing, multi-label annotation handling, and radiological bounding-box localization.

Hate Speech Detection — Transformer Fine-tuning | *Python, BERT, HuggingFace, NLTK*

- End-to-end NLP pipeline: custom merged dataset, class-imbalance handling, data augmentation, and BERT/variant fine-tuning for toxic content classification.

EDUCATION

M.Tech in Data Science

Pandit Deendayal Energy University

GPA: 9.61/10 · **Gold Medalist** - Top graduate of the program

Jul 2023 – May 2025

Gandhinagar, IN

B.E. in Computer Science & Engineering

Babaria Institute of Technology (GTU)

GPA: 9.02/10

Jul 2019 – May 2023

Vadodara, IN

PUBLICATIONS & RESEARCH

- **Lad, D. et al.** "FiLMer: Resolution and Domain-Invariant Deep Learning Emulation for Regional Weather Forecasting" *Submitted to KDD 2026, AI for Science Track* (under review).

CERTIFICATIONS & ACHIEVEMENTS

- **IBM Generative AI Engineering with LLMs Professional Certificate:** 7-course series covering LLM architectures, fine-tuning, data preparation, and production GenAI workflows.
- **Google Cloud Skills Boost (30-Day Program):** Proficiency in cloud computing, data engineering, and ML on GCP (BigQuery, Vertex AI).
- **Accenture Consultant Virtual Experience:** Exposure to consulting workflows and enterprise solution design.
- **Looker Studio (Advanced):** End-to-end dashboard design, data storytelling, and stakeholder-ready reporting.

LEADERSHIP & COMMUNITY

- **IEEE Computer Society, PDEU** - Vice Chair → Chair → Advisor (2023–2025): Led technical events, workshops, and community initiatives for student members.
- **Google Developer Student Club (GDSC), BIT** - Design Team (2021–2022).
- **National Service Scheme (NSS), GTU** - Volunteer (2020–2023).